STUDY GUIDE & LEXICON

The Epistemic Theater

How science knows what it knows, and what counts as science: a companion to the narrated program.

48:59 running time **14** chapters **134** lexicon terms **148** references

I	Aims of Science	1:37
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Each chapter lists the key positions and who holds them, the predation case files the chapter uses (anchored on Wilmers et al. 2025), formulas where they matter, every numbered lexicon term from the video with the time it appears, and practice prompts. For each prompt, separate what was observed, what was inferred, and what can be claimed. Timestamps match the video and the interactive player.

I Aims of Science

Explain, predict, understand, or intervene?

video 1:37–5:37

1:44 Case file: the Northern Range elk · 2:44 What is an explanation? · 3:36 True, or merely useful? · 4:31 Observe, infer, claim

KEY POSITIONS

IDEA	WHO	IN ONE LINE
Deductive-nomological model	Hempel & Oppenheim 1948	Explanation is deduction from laws plus conditions; explanation and prediction share one form.
Explanatory asymmetry	Bromberger 1966	The flagpole's shadow lets you deduce its height but does not explain it.
Causal-mechanical explanation	Salmon 1984	To explain is to show the causal processes and mechanisms that produced the event.
Unification	Kitcher 1989	To explain is to fit many phenomena under few argument patterns.
No-miracles argument	Putnam 1975	Realism is the only view that does not make the success of science a miracle.
Pessimistic meta-induction	Laudan 1981	Many successful theories were false, so success is weak evidence of truth.
Constructive empiricism	van Fraassen 1980	Science aims at empirical adequacy, truth about observables.
Structural realism	Worrall 1989	What survives theory change is mathematical structure.
Toulmin model	Toulmin 1958	Data support a claim through a warrant with backing, a qualifier, and possible rebuttals.

PREDATION CASE FILES

STUDY	SYSTEM	WHAT IT SHOWS
White et al. 2024	Northern Range elk	>19,000 just before 1995; 3,915 in 2013; 7,000–9,000 now.

STUDY	SYSTEM	WHAT IT SHOWS
Clark & Hebblewhite 2021	52 predator-removal experiments	Removals often raised juvenile recruitment but rarely adult survival or abundance.
Vucetich et al. 2005	Northern Range elk, 1961–2004	First post-reintroduction decade driven mainly by hunting and drought.

LEXICON

No. 001 2:31

Aims of science

Goals that inquiry pursues: description, explanation, prediction, understanding, and intervention. They can conflict.

No. 002 2:45

Deductive-nomological model

Hempel and Oppenheim's account: an explanation is a valid deduction of the event from laws of nature plus initial conditions.

No. 003 3:17

Causal-mechanical explanation

Salmon's account: to explain is to describe the causal processes and mechanisms that produced the phenomenon.

No. 004 3:17

Unification

Kitcher's account: explanation fits many phenomena under a small number of argument patterns.

No. 005 3:37

Scientific realism

The view that successful theories are approximately true, including their claims about unobservable entities.

No. 006 3:37

No-miracles argument

Realism best explains the predictive success of science; otherwise that success would be a miracle (Putnam).

No. 007 3:48

Pessimistic meta-induction

Many successful past theories were false, so success is weak evidence of truth for current ones (Laudan).

No. 008 4:01

Constructive empiricism

Van Fraassen's view that science aims only at empirical adequacy: truth about observables.

No. 009 4:32

Observation

A record produced by senses or instruments; the data a claim rests on. Never wholly free of theory (Chapter VI).

No. 010 4:32

Inference

Reasoning from observations to conclusions through models and assumptions.

No. 011 4:32

Knowledge claim

A statement offered as knowledge, with explicit scope and confidence, open to scrutiny.

No. 012 4:50

Warrant (Toulmin)

The general rule licensing the step from data to claim, supported by backing and limited by qualifiers and rebuttals.

EXAM PROMPTS

1. Distinguish explanation from prediction. Give an ecological model that predicts without explaining, and one that explains but predicts poorly.

2. State the no-miracles argument and the pessimistic meta-induction. Which position do you hold about “predation risk” or “carrying capacity”?
3. Lay out the claim “wolves caused the Northern Range elk decline” as a Toulmin argument, including a qualifier and a rebuttal.

II Knowing Nature

What is knowledge, and how do we seek it?

video 5:37–9:55

5:45 Knowledge and justification · 6:50 Four questions · 7:37 Positivism and postpositivism · 8:32 Interpretivism · 9:23 One herd, two lenses

KEY POSITIONS

IDEA	WHO	IN ONE LINE
Justified true belief	Classical analysis	Knowledge as belief that is true and held for good reasons.
Gettier problem	Gettier 1963	Justified true beliefs that are true by luck are not knowledge.
Foundationalism vs coherentism	—	Justification rests on basic beliefs, or on mutual support among beliefs.
Problem of induction	Hume 1748	Past experience cannot logically guarantee the future.
Fallibilism	Peirce 1877	Any belief may be wrong; inquiry stays open to correction.
Naturalized epistemology	Quine 1969	Epistemology is continuous with science: the study of how knowers actually learn.
Research paradigm	Guba & Lincoln 1994, 2005	Answers to ontology, epistemology, methodology, and axiology.
Positivism; verification principle	Comte; Vienna Circle; Ayer 1936	Knowledge from observation; meaningful only if verifiable or analytic. Self-undermining.
Postpositivism; critical realism	Bhaskar 1975	A real world, imperfectly known through fallible, probabilistic inquiry.
Interpretivism; Verstehen	Dilthey 1883; Weber 1922; Taylor 1971	Human action is understood through meanings, not only explained by causes.
Social construction	Berger & Luckmann 1966	Many social categories and institutions are created and sustained by people.

PREDATION CASE FILES

STUDY	SYSTEM	WHAT IT SHOWS
Haggerty & Travis 2006	Elk on private land outside Yellowstone	Tolerance of elk on irrigated fields shaped herd distribution: meanings shape management.
Wilmers et al. 2025	Social carrying capacity	The number of carnivores people tolerate is often below ecological carrying capacity.

LEXICON

No. 013 5:46

Epistemology

The study of knowledge: its nature, sources, justification, and limits.

No. 014 5:46

Justified true belief

The classical analysis of knowledge: a belief that is true and held for good reasons.

No. 015 5:58

Gettier problem

Cases of justified true belief that are true by luck, showing the classical analysis is incomplete.

No. 016 6:15

Foundationalism vs coherentism

Justification rests on basic, self-justifying beliefs, or on the mutual support of a whole system of beliefs.

No. 017 6:33

Fallibilism

Any belief, however well supported, might be mistaken and remains open to revision (Peirce).

No. 018 6:51

Ontology

Assumptions about what exists and what kind of reality is being studied.

No. 019 6:51

Methodology

The logic of how inquiry should proceed, as distinct from particular methods or techniques.

No. 020 6:51

Axiology

The values and ethical commitments of a research tradition, including what researchers owe to others.

No. 021 7:07

Research paradigm

A tradition's combined answers to questions of ontology, epistemology, methodology, and axiology (Guba and Lincoln).

No. 022 7:38

Positivism

Comte's doctrine that genuine knowledge comes from observation and the laws summarizing it, not metaphysics.

No. 023 7:38

Verification principle

Logical positivism's criterion: a statement is meaningful only if empirically verifiable or true by definition.

No. 024 8:12

Postpositivism

The fallibilist successor to positivism: a real world, known imperfectly, approached through design, statistics, and replication.

No. 025 8:12

Critical realism

The view that a mind-independent reality exists and can be known, though only partially and through fallible theory.

No. 026 8:33

Interpretivism

The tradition holding that social life must be understood through the meanings people give it, not only explained by causes.

No. 027 8:33

Verstehen

Interpretive understanding of action from the actor's point of view (Dilthey, Weber); contrasted with causal explanation, Erklären.

No. 028 8:49

Social construction

The process by which categories and institutions are created and sustained by people (Berger and Luckmann).

No. 029 9:06

Social carrying capacity

The population size of a species that people will tolerate, often below ecological carrying capacity.

EXAM PROMPTS

1. Construct an ecological Gettier case of your own and say what it shows about justification.
2. Compare postpositivism and interpretivism on ontology, epistemology, methodology, and axiology.
3. Why did the verification principle fail, and what replaced it in practice?

III Inference

Which inferences preserve truth, and which expand knowledge?

video 9:55–12:44

10:02 Case file: the Kaibab deer · 10:47 Three kinds of inference · 11:49 Strong inference

KEY POSITIONS

IDEA	WHO	IN ONE LINE
Deduction	—	Truth-preserving, not ampliative. Valid if the conclusion must follow; sound if valid with true premises.
Induction	Hume 1748	Ampliative and fallible; cannot be justified without assuming nature is uniform.
Abduction / IBE	Peirce 1878; Lipton 2004	Infer the hypothesis that would best explain a surprising fact; “loveliest” is not always “likeliest”.
Hypothetico-deductive method	—	Deduce observable consequences of a hypothesis and test them.
Multiple working hypotheses	Chamberlin 1890	Hold rivals simultaneously to avoid attachment to one.
Strong inference	Platt 1964	Devise rival hypotheses and crucial experiments that exclude some; iterate.

PREDATION CASE FILES

STUDY	SYSTEM	WHAT IT SHOWS
Leopold 1949	Kaibab Plateau deer	Deer irruption and range collapse attributed to predator removal.
Hairston, Smith & Slobodkin 1960	Green world hypothesis	The world is green because predators hold herbivores in check.
Caughley 1970	Kaibab records	Data “unreliable and inconsistent”; cattle removal an untested rival.
Ford & Goheen 2015	Trophic-cascade research	Call for strong inference and mechanism.

LEXICON

No. 030 10:18

Green world hypothesis

Hairston, Smith and Slobodkin (1960): the world is green because predators hold herbivores below levels that would strip vegetation.

No. 031 10:18

Trophic cascade

Indirect effects of predators that propagate down a food chain, for example to plants by suppressing herbivores.

No. 032 10:48

Deduction

Inference in which true premises guarantee a true conclusion; truth-preserving but not ampliative.

No. 033 10:48

Validity and soundness

An argument is valid if its conclusion must follow from its premises, and sound if it is valid and its premises are true.

No. 034 11:05

Induction

Ampliative inference from observed cases to general conclusions or unobserved cases; fallible.

No. 035 11:05

Problem of induction

Hume's argument that induction cannot be justified without circularly assuming that nature is uniform.

No. 036 11:25

Abduction

Inference from a surprising observation to the hypothesis that would best explain it (Peirce); inference to the best explanation (Lipton).

No. 037 11:50

Hypothetico-deductive method

Propose a hypothesis, deduce observable consequences, and test them; failure counts against the hypothesis.

No. 038 12:21

Multiple working hypotheses

Chamberlin's method of holding several rival explanations at once to avoid premature attachment to one.

No. 039 12:21

Strong inference

Platt's procedure: devise alternative hypotheses, design crucial experiments that exclude some, and iterate.

EXAM PROMPTS

1. Classify each step of the Kaibab story as deduction, induction, or abduction, and identify the untested rival.
2. Derive opposing predictions for fear-mediated versus density-mediated aspen recovery.
3. What does Platt's strong inference add to the hypothetico-deductive method?

IV Falsification

Can a single failed prediction refute a theory?

video 12:44–15:45

12:50 Popper's asymmetry · 13:31 Case file: does fear lower pregnancy? · 14:23 Duhem, Quine, and the protective belt

KEY POSITIONS

IDEA	WHO	IN ONE LINE
Logical asymmetry	Popper 1959	No number of confirmations proves a universal claim; one counterinstance refutes it.
Corroboration	Popper 1963	Survival of severe tests; not proof or probability.
Falsifiability as demarcation	Popper 1963	Science makes risky predictions; theories that explain anything explain nothing.
Duhem–Quine thesis	Duhem 1906; Quine 1951	Hypotheses are tested with auxiliaries, so a failure does not say what is false.
Research programmes	Lakatos 1970	Hard core, protective belt; progressive if adjustments predict novel facts.
NHST is not falsification	Johnson 1999; Anderson et al. 2000	Rejecting a null of no effect is not a risky test of a research hypothesis.

PREDATION CASE FILES

STUDY	SYSTEM	WHAT IT SHOWS
Creel et al. 2007	Elk progesterone vs elk:wolf ratio	Reported lower progesterone where wolves were relatively more numerous.
White et al. 2011; Proffitt et al. 2014	Direct pregnancy measurements	Pregnancy uniformly high regardless of wolf risk.
Middleton et al. 2013	GPS-collared elk and wolves	Encounters too infrequent to disrupt foraging.
Kauffman et al. 2010; Beschta & Ripple 2013	Risk and aspen browsing	No risk effect found; reply: elk sense risk where they meet wolves, not where wolves kill.

LEXICON

No. 040 12:51

Logical asymmetry

No finite set of confirming instances proves a universal claim, but one genuine counterinstance refutes it.

No. 041 13:03

Corroboration

Popper's term for a theory's record of surviving severe attempts at refutation; not proof or probability.

No. 042 13:21

Falsifiability

The property of a theory that some possible observation would contradict it; Popper's mark of science.

No. 043 13:32

Nonconsumptive effects

Effects of predators on prey through fear: changes in behavior or physiology that may reduce fitness without killing.

No. 044 14:04

Auxiliary hypothesis

An assumption, such as how an instrument or index works, needed alongside the main hypothesis to derive a prediction.

No. 045 14:24

Duhem–Quine thesis

Hypotheses are tested only in conjunction with auxiliary assumptions, so a failed prediction cannot by itself identify what is false.

No. 046 14:56

Research program

Lakatos: a hard core of central commitments, shielded by a protective belt of revisable auxiliary hypotheses.

No. 047 14:56

Progressive vs degenerating

A program is progressive when its adjustments predict novel facts that are confirmed, degenerating when they only accommodate failures.

No. 048 15:29

Null hypothesis significance testing

Testing whether data are improbable under a null model, usually of no effect; rejecting it does not test a substantive hypothesis severely.

EXAM PROMPTS

1. Is the hypothesis that fear lowers elk pregnancy falsified? Identify the auxiliaries and say what Lakatos would ask.
2. Explain why Neptune and Vulcan illustrate the Duhem–Quine problem.
3. Why is rejecting a null hypothesis not a Popperian test? What would a severe test of a cascade look like?

V Paradigms

How does science change its mind?

video 15:45–18:27

15:51 Normal science and revolution · 16:39 Case file: from vermin to keystone · 17:44 Choosing between theories

KEY POSITIONS

IDEA	WHO	IN ONE LINE
Normal science; puzzle-solving	Kuhn 1962	Research within a shared paradigm; anomalies set aside until crisis.
Disciplinary matrix; exemplar	Kuhn 1970	The two senses of paradigm: shared commitments, and model solutions students imitate.
Incommensurability; gestalt switch	Kuhn 1962	Rival paradigms lack a common measure; change resembles seeing the duck–rabbit flip.
Theory-choice values	Kuhn 1977	Accuracy, consistency, scope, simplicity, fruitfulness, weighed differently.
Epistemological anarchism	Feyerabend 1975	“Anything goes”: no method survives the history of science intact.
Research traditions	Laudan 1977	Traditions judged by problem-solving success.
Balance of nature to patch dynamics	Wu & Loucks 1995	A candidate paradigm shift in ecology.

PREDATION CASE FILES

STUDY	SYSTEM	WHAT IT SHOWS
Leopold 1949	“Thinking Like a Mountain”	A conversion from predator control to predator appreciation.
Estes et al. 1998; Terborgh et al. 2001; Crooks & Soulé 1999	Exemplars of top-down control	Otters–kelp; predator-free islands; mesopredator release and songbirds.
Ripple & Beschta 2003, 2006, 2012	Yellowstone cascade	Normal science extending the pattern to

STUDY	SYSTEM	WHAT IT SHOWS
		cottonwood, willow, and aspen.

LEXICON

No. 049 16:03

Paradigm (Kuhn)

A community's shared disciplinary matrix of theory, methods, and values, anchored by exemplary problem solutions.

No. 050 16:03

Exemplar

A concrete, model problem solution that students learn to imitate and that defines a paradigm in practice.

No. 051 16:03

Normal science

Puzzle-solving research within an accepted paradigm, which sets aside anomalies rather than testing the paradigm itself.

No. 052 16:21

Gestalt switch

Kuhn's image for a paradigm change: the same data seen as a different whole, like the duck–rabbit figure.

No. 053 16:21

Incommensurability

Rival paradigms lack a common measure: they differ in problems, standards, and meanings, so no neutral test decides.

No. 054 17:09

Mesopredator release

An increase in mid-sized predators after apex predators decline, through reduced predation or competition.

No. 055 17:45

Theory-choice values

Kuhn's shared criteria of accuracy, consistency, scope, simplicity, and fruitfulness, weighed differently by different scientists.

No. 056 18:02

Epistemological anarchism

Feyerabend's thesis that no fixed methodological rule holds across the history of science: "anything goes."

EXAM PROMPTS

1. Is the move from predator control to trophic cascades a Kuhnian revolution? Use exemplars and anomalies in your answer.
2. Distinguish Kuhn's paradigm from a research paradigm in Guba and Lincoln's sense.
3. How can theory choice be rational if paradigms are incommensurable?

VI Observation

Is there such a thing as a pure observation?

video 18:27–21:48

18:33 Seeing as · 19:11 Case file: the five tallest stems · 20:16 Instruments and definitions · 21:22 Laden, not determined

KEY POSITIONS

IDEA	WHO	IN ONE LINE
Theory-ladenness; seeing-as	Hanson 1958	What is observed depends on concepts brought to it (Tycho and Kepler at dawn).
Data vs phenomena	Bogen & Woodward 1988	Theories explain stable phenomena inferred from data, not the data themselves.
Triangulation	Hacking 1983	Agreement among independent methods built on different theories.
Imperfect detection	MacKenzie et al. 2002	Counts confound abundance with detection; non-detection ≠ absence.
Reflexivity	Interpretive research	Declaring the researcher's position because it shapes what is noticed.

FORMULAS

Expected count	$\mathrm{E}[C] = N \cdot p$ (abundance × detection probability)
Occupancy, one site	$\mathrm{P}(\text{detected at least once in } K \text{ surveys}) = \psi[1 - (1 - p)^K]$

PREDATION CASE FILES

STUDY	SYSTEM	WHAT IT SHOWS
Ripple & Beschta 2007	Five tallest aspen per stand	Designed to detect early recovery among the leading stems.
Painter et al. 2015	Random plots and the five tallest stems, 2012	Up to 65% of stands had a sapling above 200 cm; the range-wide inference rested largely on the random samples.

STUDY	SYSTEM	WHAT IT SHOWS
Kauffman et al. 2010	All young aspen in random plots	No difference with predation risk.
Brice et al. 2022, 2024	Both designs on the same stands	Both detect a cascade; tallest stems show whether, random samples show how strong.
Hobbs et al. 2024	Willow sampling	Studies at well-watered sites overstated recovery.
Kohl et al. 2018, 2019	GPS on elk, wolves, pumas	Diel landscape of fear invisible to sparse VHF data.

LEXICON

No. 057 18:51

Theory-ladenness

The dependence of what is observed, described, and recorded on the observer's concepts, expectations, and instruments (Hanson).

No. 058 18:51

Data vs phenomena

Data are particular records; phenomena are stable features inferred from them. Theories explain phenomena (Bogen and Woodward).

No. 059 19:49

Sampling design

The rule for choosing which units to measure. It determines which population, and which question, the data can answer.

No. 060 20:03

Sampling bias

Systematic difference between the sampled units and the population of interest, which distorts estimates.

No. 061 20:34

Landscape of fear

The spatial and temporal pattern of predation risk perceived by prey, which can shape where and when they forage.

No. 062 20:51

Operational definition

A definition of a concept in terms of the procedure used to measure it, such as "risk" as distance to a kill site.

No. 063 21:06

Imperfect detection

Animals present may go unrecorded, so counts confound abundance with detection probability (MacKenzie et al.).

No. 064 21:23

Triangulation

Agreement among independent methods with different assumptions, which reduces the chance a result is an artifact (Hacking).

No. 065 21:39

Reflexivity

Critical attention to how the researcher's position, values, and presence shape the research.

EXAM PROMPTS

1. Explain how the tallest-stem and random-plot designs each embody a question, and which question each can answer.
2. Give three operational definitions of “predation risk” and the theory each carries.
3. Does theory-ladenness mean anything goes? Answer with Hacking’s triangulation.

VII Models

How can a false model teach us something true?

video 21:48–24:25

21:54 The map and the territory · 22:49 Case file: the three-box cascade · 23:40 Models that act

KEY POSITIONS

IDEA	WHO	IN ONE LINE
All models are wrong	Box 1976	Models are useful because they omit; the question is whether they are wrong in ways that matter.
Three kinds of idealization	Weisberg 2007	Galilean (tractability), minimalist (difference-makers), multiple-models.
Models as mediators	Morgan & Morrison 1999	Partly independent of theory and world, so they can teach about both.
Generality, realism, precision	Levins 1966	No model maximizes all three; robustness analysis across models.
Estimand vs estimate	—	The quantity of interest versus the model output; the link is an assumption.

PREDATION CASE FILES

STUDY	SYSTEM	WHAT IT SHOWS
Metz et al. 2020	Pumas vs wolves, 1998–2004	Pumas killed a larger share of Northern Range elk than wolves in most years.
Barber-Meyer et al. 2008	Elk calves	Bears were the main predators of calves.
Ripple et al. 2025 vs a critique cited in Wilmers et al. 2025	Willow crown volume	+1,500% estimate vs concern that the volume model was calibrated on lightly browsed plants.
Messier 1994; National Research Council 1997	Predator pit	Little support for alternative stable states in wolf–moose systems.

LEXICON

No. 066 22:08

Idealization

Deliberate distortion or omission in a model to make it tractable or to isolate the factors that matter (Weisberg).

No. 067 22:08

Models as mediators

Models are partly independent of both theory and data, so they can be used to learn about each (Morgan and Morrison).

No. 068 22:27

Generality–realism–precision

Levins's tradeoff: a model can maximize at most two of generality, realism, and precision.

No. 069 22:27

Robustness analysis

Checking whether a result holds across models with different simplifying assumptions.

No. 070 23:19

Extrapolation

Applying a model outside the range of conditions used to build or calibrate it.

No. 071 23:19

Estimand vs estimate

The estimand is the quantity a study aims to know; the estimate is what a model and data produce. The link between them is an assumption.

No. 072 23:41

Predator pit

A hypothesized low-density prey equilibrium maintained by predation, separated from a high-density equilibrium.

No. 073 23:58

Alternative stable states

Different self-reinforcing states a system can occupy under the same conditions; shifting between them requires a large perturbation.

EXAM PROMPTS

1. What is wrong, and what is right, about the three-box wolf–elk–aspen cascade?
2. Explain extrapolation using the willow crown-volume dispute.
3. How did the predator-pit model act in the world, and what did tests find?

VIII Causality

What would have happened otherwise?

video 24:25–28:55

24:32 Association is not explanation · 25:31 Would that elk have died anyway? · 26:38 Drawing the causes · 27:37 Designs that break the symmetry

KEY POSITIONS

IDEA	WHO	IN ONE LINE
Confounding; reverse causation; collider bias	—	Three non-causal sources of association.
Simpson's paradox	—	Subgroup trends reverse when pooled; resolving it needs a causal model.
Regularity, counterfactual, interventionist	Hume 1748; Lewis 1973; Woodward 2003	Three accounts of what a cause is.
Potential outcomes	Rubin 1974; Holland 1986	Each unit has outcomes with and without treatment; only one is observed.
DAGs; back-door criterion; ladder	Pearl 2009; Pearl & Mackenzie 2018	Block non-causal paths; association, intervention, counterfactuals.
Randomization; factorial designs	Fisher 1935	Balance confounders in expectation; reveal interactions.
BACI; Hill's viewpoints	Stewart-Oaten et al. 1986; Hill 1965	Weighing observational evidence when experiments are impossible.

FORMULAS

Individual causal effect $\tau_i = Y_i(1) - Y_i(0)$ (never observed directly)

Average causal effect $\mathrm{ATE} = \mathrm{E}[Y(1)] - \mathrm{E}[Y(0)]$, identified by randomization

Back-door adjustment $\mathrm{P}(y \mid \mathrm{do}(x)) = \Sigma_z \, \mathrm{P}(y \mid x, z) \, \mathrm{P}(z)$

PREDATION CASE FILES

STUDY	SYSTEM	WHAT IT SHOWS
Brodie et al. 2013	45 elk populations	Higher survival with both pumas and wolves, most likely because hunting quotas were cut; harvest dominated.
MacNulty et al. 2020	Ages of killed cow elk	Wolf kills modally ~16 years old; hunter kills ~4: compensatory vs additive.
Ford & Goheen 2015; Wilmers et al. 2025	Three causal links of a cascade	Herbivory–plants established by exclosures; predators–herbivores least certain; net effect mixed.
Bilyeu et al. 2008; Hobbs et al. 2024	Dams × exclosures	Together, willows passed browse height in ~5 years; controls never did.
Hurley et al. 2011; Bishop et al. 2009	Mule deer	Predator removal barely changed growth; winter feeding raised it 13% per year.
Wilmers et al. 2006	Isle Royale parvovirus	Wolf crash as a natural experiment; moose grew faster and higher.

LEXICON

No. 074 24:33

Confounding

A common cause of exposure and outcome that creates or distorts an association between them.

No. 075 24:33

Collider bias

A spurious association created by conditioning or selecting on a variable caused by both exposure and outcome.

No. 076 25:18

Simpson's paradox

An association within every subgroup that reverses when the subgroups are combined.

No. 077 25:32

Counterfactual

A claim about what would have happened had some condition been different; the basis of Lewis's analysis of causation.

No. 078 25:32

Interventionist causation

C causes E if an intervention changing C, holding other things fixed, would change E (Woodward).

No. 079 25:50

Fundamental problem of causal inference

Each unit reveals only one of its potential outcomes, so individual causal effects are never directly observed (Holland).

No. 080 26:05

Additive vs compensatory mortality

Additive deaths raise total mortality; compensatory deaths replace deaths that would have occurred from other causes.

No. 081 26:39

Directed acyclic graph

A diagram of causal assumptions in which arrows point from causes to effects and no path loops back on itself.

No. 082 26:57

Back-door criterion

Pearl's rule for which variables to adjust for: block every non-causal path that enters the exposure through an arrow pointing into it.

No. 083 26:57

Ladder of causation

Pearl's three levels of causal questions: association (seeing), intervention (doing), and counterfactuals (imagining).

No. 084 27:38

Randomization

Assigning treatments by chance so that confounders are balanced across groups in expectation (Fisher).

No. 085 27:38

Interaction

When the effect of one factor depends on the level of another, revealed by factorial designs.

No. 086 28:21

Natural experiment

A comparison created by events outside the researcher's control that approximates random assignment.

No. 087 28:38

BACI design

Before-after-control-impact: compares change at impact sites with change at control sites over the same period.

EXAM PROMPTS

1. Draw a DAG for the Northern Range and identify a back-door path between wolves and elk survival.
2. Why is compensatory mortality a counterfactual claim? What evidence bears on it?
3. Design a study that separates top-down from bottom-up limitation of a deer population.

IX Underdetermination

What if the evidence fits more than one theory?

video 28:55–31:49

29:02 One body of evidence, many theories · 29:50 Case file: one decline, many causes · 31:01 Breaking the tie

KEY POSITIONS

IDEA	WHO	IN ONE LINE
Underdetermination	Duhem 1906; Quine 1951	Evidence may not fix which theory is true.
Transient vs strong	—	Resolvable by further evidence, or empirically equivalent forever.
Equifinality	Beven & Freer 2001	Many parameter sets fit the same record.
Pattern–process problem	Levin 1992	Different processes generate the same pattern.
Multimodel inference	Burnham & Anderson 2002	Rank and average candidate models by information criteria.
Scale dependence	Prugh & Sivy 2020; Sivy et al. 2017	Positive associations locally, negative across landscapes.

FORMULAS

AIC

$$\mathrm{AIC} = -2 \ln L + 2k; \quad \Delta_i = \mathrm{AIC}_i - \mathrm{AIC}_{\min}$$

Akaike weight

$$w_i = \exp(-\Delta_i/2) \,/\, \Sigma_j \exp(-\Delta_j/2)$$

PREDATION CASE FILES

STUDY	SYSTEM	WHAT IT SHOWS
Vucetich et al. 2005	Elk decline, first decade	Hunting and drought; no comparable apportionment since.
MacNulty et al. 2020	Winter range outside the park (20% → 80%)	Demographic, not behavioural: survival was higher outside.
Brice et al. 2024	Aspen recruitment	Risk effects 8–28 times weaker than density effects.

STUDY	SYSTEM	WHAT IT SHOWS
Hobbs et al. 2024 vs Ripple et al. 2025	Willow	Stalled system (possible alternative stable state) vs slow recovery, from the same data.

LEXICON

No. 088 29:03

Underdetermination

The evidence available is insufficient to determine which of several rival theories is true.

No. 089 29:19

Transient vs strong underdetermination

Transient: resolvable by further evidence. Strong: rivals are empirically equivalent, so no possible evidence decides.

No. 090 29:35

Equifinality

Many different model structures or parameter sets reproduce the same observations equally well (Beven).

No. 091 29:35

Pattern–process problem

Different processes can generate the same pattern, so patterns alone rarely identify their cause (Levin).

No. 092 31:02

Theoretical virtues

Non-empirical criteria such as simplicity, scope, fruitfulness, and coherence used to choose among theories that fit the data.

No. 093 31:13

Multimodel inference

Ranking and averaging a set of candidate models by information criteria such as AIC, rather than selecting one (Burnham and Anderson).

No. 094 31:30

Scale dependence

The direction or strength of an ecological relationship changes with the spatial or temporal scale of study (Wiens; Levin).

EXAM PROMPTS

1. Which parts of the Northern Range story show transient underdetermination that has been resolved, and which remain open?
2. Explain how scale can make two contradictory carnivore studies both correct.
3. What does multimodel inference do that a single best model does not?

X Bayesian Evidence

How much should evidence change belief?

video 31:49–35:15

31:56 Degrees of belief · 32:43 The strength of evidence · 33:34 Case file: updating on the cascade · 34:45 Priors and their critics

KEY POSITIONS

IDEA	WHO	IN ONE LINE
Credence; Dutch book	Ramsey 1926	Degrees of belief should obey probability, or a set of bets loses for sure.
Conditionalization; confirmation	—	Update to P(H | E); evidence confirms by raising probability.
Likelihood ratio	Royall 1997	The strength of evidence; Royall's three questions: evidence, belief, action.
Base-rate fallacy	—	Ignoring how rare the hypothesis is.
Washing out of priors	—	Different reasonable priors converge with enough shared data.
Severe testing	Mayo 2018	Support only when a claim passed a test it would likely have failed if false.

FORMULAS

Bayes' theorem

$$\mathrm{P}(H \mid E) = \mathrm{P}(E \mid H)\,\mathrm{P}(H) \,/\, \mathrm{P}(E)$$

Odds form

$$\text{posterior odds} = \text{likelihood ratio} \times \text{prior odds}; \quad \mathrm{LR} = \mathrm{P}(E \mid H_1) \,/\, \mathrm{P}(E \mid H_2)$$

Camera-trap example (illustrative)

$$\mathrm{P}(\text{wolf} \mid \text{flag}) = (0.95 \times 0.001) \,/\, (0.95 \times 0.001 + 0.01 \times 0.999) \approx 0.087$$

Flat-prior reading of a CI (illustrative)

$$\mathrm{SE} \approx (0.05 + 0.41)/3.92 \approx 0.117; \quad \mathrm{P}(\text{effect} < 0) \approx \Phi(0.18/0.117) \approx 0.94$$

PREDATION CASE FILES

STUDY	SYSTEM	WHAT IT SHOWS
McLaren & Peterson 1994	Isle Royale tree rings	Fir growth tracked wolves and moose: a prior for cascades.
Kauffman 2010 → Hobbs 2024	Yellowstone studies in sequence	Belief in some aspen cascade survived; belief that fear drives it fell.
Wilmers et al. 2025	16 large carnivore–mesocarnivore pairs	Mean correlation −0.18 (95% CI −0.41 to 0.05; p = 0.11).

LEXICON

No. 095 31:57

Credence

A degree of belief, represented as a probability between zero and one.

No. 096 31:57

Dutch book argument

Credences that violate the probability axioms expose you to a set of bets that loses in every outcome (Ramsey).

No. 097 32:16

Bayes' theorem

P(H | E) = P(E | H) P(H) / P(E): the posterior is proportional to the prior times the likelihood.

No. 098 32:30

Conditionalization

Updating by adopting, as the new credence in H, the old conditional probability of H given the evidence.

No. 099 32:44

Likelihood ratio

P(E | H1) / P(E | H2): how much more probable the evidence is under one hypothesis than another; the measure of evidential strength.

No. 100 32:58

Base-rate fallacy

Judging a hypothesis by the accuracy of a test while ignoring how rare the condition is to begin with.

No. 101 34:46

Washing out of priors

With enough shared evidence, posteriors from different reasonable priors converge.

No. 102 35:00

Severe testing

Mayo's error-statistical criterion: data support a claim only if the claim passed a test it would probably have failed were it false.

EXAM PROMPTS

1. Work the camera-trap base-rate example and explain why most flagged "wolves" are coyotes.
2. What does a non-significant result with interval −0.41 to 0.05 not show? Answer in Bayesian and in error-statistical terms.
3. How should a prior for the strength of the Yellowstone cascade change across the studies in this chapter?

XI Replication

Would it happen again, and elsewhere?

video 35:15–38:47

35:23 Four ways to hold up · 35:56 The replication crisis · 36:56 Case file: thirty kilometers apart · 38:13 Remedies and other standards

KEY POSITIONS

IDEA	WHO	IN ONE LINE
Reproducibility; replicability	National Academies 2019	Same data and code vs new data; plus generalizability and robustness.
Many analysts	Silberzahn et al. 2018	29 teams, one data set, estimates from slightly negative to strongly positive.
Replication crisis	Ioannidis 2005; Open Science Collaboration 2015	97% of originals significant; 36% of replications.
P-hacking; HARKing; forking paths	Kerr 1998; Gelman & Loken 2014	Flexible analysis inflates false positives.
Pseudoreplication	Hurlbert 1984	Non-independent measurements treated as replicates.
Preregistration; transferability	Nosek et al. 2018; Yates et al. 2018	Fix plans before data; test models in new places.
Trustworthiness criteria	Lincoln & Guba 1985	Credibility, transferability, dependability, confirmability.

PREDATION CASE FILES

STUDY	SYSTEM	WHAT IT SHOWS
Garrott et al. 2008	Madison Headwaters elk	Cow survival down 5–15%; numbers down 5- to 10-fold.
Klein 1995; Roffler et al. 2023	Coronation and Pleasant islands	Wolves nearly eliminated deer.
Griffin et al. 2011	11 elk populations, neonates	Only grizzly predation additive.

STUDY	SYSTEM	WHAT IT SHOWS
Brodie et al. 2013	45 populations, adult females	Human harvest dominated.
Prugh & Sivy 2020; Wilmers et al. 2025	Where studies are vs where carnivores live	~70% of interaction studies in protected areas; 15% of range protected, 4% strictly.

LEXICON

No. 103 35:24

Reproducibility

Obtaining consistent results from the same data using the same methods and code.

No. 104 35:24

Replicability

Obtaining consistent results across studies that address the same question with new data.

No. 105 35:38

Generalizability

The extent to which results apply in contexts, places, or populations beyond those studied.

No. 106 35:38

Robustness

Whether a result survives reasonable alternative analyses of the same data.

No. 107 36:18

Publication bias

Selective publication of positive or significant results, which inflates effects in the literature.

No. 108 36:18

P-hacking

Trying analyses, subsets, or stopping rules until a result crosses a significance threshold.

No. 109 36:18

HARKing

Hypothesizing after the results are known and presenting the hypothesis as if it had been made in advance (Kerr).

No. 110 36:18

Garden of forking paths

Data-dependent analytic choices that inflate false positives even without deliberate fishing (Gelman and Loken).

No. 111 36:40

Pseudoreplication

Treating non-independent measurements, such as repeated samples of one experimental unit, as independent replicates (Hurlbert).

No. 112 37:44

External validity

Whether a result holds beyond the conditions, sites, or populations in which it was obtained.

No. 113 37:59

Scope conditions

The circumstances under which a generalization is expected to hold.

No. 114 37:59

Apparent competition

Two prey share a predator; one prey sustains more predators, which then depress the other (Holt).

No. 115 38:14

Preregistration

Recording hypotheses, design, and analysis plan before data are collected or examined.

No. 116 38:33

Trustworthiness criteria

Lincoln and Guba’s standards for interpretive research: credibility, transferability, dependability, and confirmability.

EXAM PROMPTS

1. Distinguish reproducibility, replicability, generalizability, and robustness with one ecological example each.
2. Why did wolves limit elk in the Madison Headwaters but not clearly on the Northern Range? State the scope conditions.
3. What does the mismatch between where carnivore studies occur and where carnivores live imply for external validity?

XII Values

Can science be value-free, and should it be?

video 38:47–42:18

38:53 The argument from inductive risk · 39:41 Case file: whose costs? · 40:38 Whose baseline? · 41:16 Objectivity as a social achievement

KEY POSITIONS

IDEA	WHO	IN ONE LINE
Value-free ideal; discovery vs justification	Reichenbach 1938	Values may guide questions and uses, not the weighing of evidence.
Argument from inductive risk	Rudner 1953; Douglas 2000, 2009	How much evidence is enough depends on the cost of error.
Epistemic vs non-epistemic values	Douglas 2009	Non-epistemic values may set thresholds but never substitute for evidence.
CUDOS norms	Merton 1942	Communalism, universalism, disinterestedness, organized skepticism.
Transformative criticism	Longino 1990	Objectivity as a property of critical communities.
Strong objectivity	Harding 1991	Starting from marginalized standpoints exposes hidden assumptions.
Crisis discipline	Soulé 1985	Conservation must act before evidence is complete; its goals are values.

PREDATION CASE FILES

STUDY	SYSTEM	WHAT IT SHOWS
Lamb et al. 2024; Serrouya et al. 2021	Mountain caribou	−51% (1991–2023); 15 of 41 subpopulations functionally extirpated; apparent competition.
Raynor et al. 2021	Wolves in Wisconsin	Deer–vehicle collisions −24%; benefits ≈ 63× livestock losses.
Gilbert et al. 2017	Puma recovery, eastern US	Projected 155 human deaths averted over 30 years.

STUDY	SYSTEM	WHAT IT SHOWS
Wilmers et al. 2025	Baselines	Disagreement over recovery partly reflects pre-European vs pre-Indigenous baselines.

LEXICON

No. 117 38:54

Value-free ideal

The view that non-epistemic values should play no role in the internal reasoning of science, only in choosing problems and applying results.

No. 118 38:54

Discovery vs justification

Reichenbach's distinction between how hypotheses arise and how they are evaluated against evidence.

No. 119 39:08

Inductive risk

The chance of error in accepting or rejecting a hypothesis; weighing its consequences requires values (Rudner, Douglas).

No. 120 39:23

Epistemic vs non-epistemic values

Values bearing on truth, such as accuracy and simplicity, versus values bearing on consequences, such as safety or livelihoods.

No. 121 40:02

Type I and Type II errors

Rejecting a true null (a false alarm) versus failing to reject a false one (a miss); in management, each carries different costs.

No. 122 40:39

Ecological baseline

The reference condition against which change or recovery is measured; choosing one is partly a value judgment.

No. 123 40:58

Crisis discipline

Soulé's description of conservation biology: a value-driven science that must act before evidence is complete.

No. 124 41:17

CUDOS norms

Merton's norms of science: communalism, universalism, disinterestedness, and organized skepticism.

No. 125 41:17

Transformative criticism

Longino's account of objectivity as a property of communities with venues for criticism, uptake, shared standards, and equality of authority.

No. 126 41:36

Strong objectivity

Harding's view that starting inquiry from marginalized standpoints exposes assumptions and so increases objectivity.

EXAM PROMPTS

1. Apply the argument from inductive risk to a decision to cull wolves for caribou.
2. Which values enter when choosing an ecological baseline for "recovery"?
3. Explain how the Yellowstone debate illustrates objectivity as a social achievement.

XIII Demarcation

What counts as science, and who decides?

video 42:18–46:05

42:26 Why the line matters · 43:05 A century of criteria · 44:21 The answer depends on the criterion · 45:15 Case file: the Kaibab story, revisited

KEY POSITIONS

IDEA	WHO	IN ONE LINE
Verifiability; falsifiability	Ayer 1936; Popper 1959	Early criteria; each fails as necessary and sufficient.
Puzzle-solving; progress	Kuhn 1962; Lakatos 1970	Criteria based on traditions and programmes.
Stagnation test	Thagard 1978	Pseudoscience stagnates while rivals progress, and its practitioners ignore that.
Methodological naturalism	McLean 1982; Kitzmiller 2005	The working rule courts applied to creationism and intelligent design.
Daubert standard	Daubert 1993	Testability, peer review, error rates, standards, general acceptance.
Demise of demarcation	Laudan 1983	No necessary and sufficient conditions survive.
Cluster concept; systematicity	Pigliucci & Boudry 2013; Hoyningen-Huene 2013	Looser revivals; science as more systematic knowledge.
Wissenschaft	—	German for all systematic knowledge, humanities included.

PREDATION CASE FILES

STUDY	SYSTEM	WHAT IT SHOWS
Leopold 1949 → Caughley 1970	The Kaibab story, revisited	A weakly tested account lived in textbooks until criticism corrected it: demarcation concerns practices.

LEXICON

No. 127 42:27

Demarcation problem

The problem of distinguishing science from non-science and pseudoscience, and of saying on what grounds.

No. 128 42:46

Daubert standard

The US test for admitting expert scientific testimony: testability, peer review, error rate, standards, and general acceptance.

No. 129 43:06

Pseudoscience

A practice presented as science that fails its standards, for example by stagnating while ignoring progressive rivals (Thagard).

No. 130 43:29

Methodological naturalism

The working rule that scientific explanations appeal only to natural causes, whatever one believes beyond them.

No. 131 44:00

Cluster concept

A concept defined by a family of overlapping features, none necessary or sufficient on its own (Pigliucci).

No. 132 44:00

Systematicity

Hoyningen-Huene's thesis that scientific knowledge differs from everyday knowledge chiefly by being more systematic.

No. 133 44:00

Wissenschaft

German for organized, systematic knowledge of any kind, including the humanities; broader than English "science".

EXAM PROMPTS

1. Apply three demarcation criteria to postpositivist ecology, interpretive human-dimensions research, and astrology. Where do the answers differ?
2. Is the label "science" a definitional choice? What follows from choosing a narrow or a broad definition?
3. Why does the Kaibab story matter for demarcation?

XIV Synthesis

How do you audit a knowledge claim?

video 46:05–48:59

46:13 Auditing a famous claim · 47:38 Seven questions for any claim · 48:19 Exam rehearsal

SEVEN-QUESTION CLAIM AUDIT

1. What was observed, and how?
2. What was inferred, under which assumptions?
3. Which alternatives remain?
4. How strong is the evidence, and what would change your mind?
5. Has it been replicated, and where does it generalize?
6. Which values and stakes shape the claim?
7. What would another tradition of inquiry ask that this one did not?

LEXICON

No. 134 47:39

Claim audit

Seven questions: what was observed and how; what was inferred and how; alternatives; strength of evidence; replication and scope; values; other traditions.

REHEARSAL ANSWERS

What separates induction from abduction, and where does strong inference fit?

Induction generalizes from observed cases to unobserved ones; abduction infers the hypothesis that would best explain a surprising fact. Strong inference disciplines abduction by generating rival explanations and designing tests that exclude some, so the loveliest story must also survive elimination.

Why does the Duhem–Quine thesis complicate falsification?

A prediction follows from a hypothesis plus auxiliary assumptions, so a failed prediction refutes only the conjunction. Logic cannot say whether the hypothesis or an auxiliary (an index, an instrument, a measure of risk) is at fault; Lakatos asks whether rescuing adjustments predict novel facts.

What did Kuhn mean by incommensurability?

Rival paradigms differ in problems, standards, and meanings of terms, so there is no neutral measure by which a single test decides between them; choice relies on shared values weighed by judgment.

Why is compensatory mortality a counterfactual claim?

It asserts that animals killed by a predator would have died anyway from other causes: a claim about an unobserved outcome. Evidence such as the age and condition of killed animals (wolf-killed cow elk ~16 years, hunter-killed ~4) bears on it indirectly.

What is the argument from inductive risk?

Because evidence never makes a hypothesis certain, accepting it requires judging the evidence sufficient, and sufficiency depends on the costs of false acceptance and false rejection. Non-epistemic values therefore legitimately set evidential thresholds (Rudner; Douglas), though they must not replace evidence.

How does the answer to "what counts as science?" change with the criterion?

Under falsifiability even astrology qualifies; under institutional norms and progress interpretive research qualifies and astrology does not; under cluster, systematicity, or Wissenschaft accounts both research traditions qualify. The practices do not change; the label does.

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Generated from the same source as the video, so term numbers and timestamps stay in sync. Narration: ElevenLabs voice “Megan” (Multilingual v2). Figures in the video marked “stylized” or “illustrative” show qualitative patterns or invented example numbers, not reproduced data. Running theme: the ecological impacts of large-carnivore recovery (Wilmers et al. 2025).